The Convex Origin of Fixed Costs[1]
Online Appendix

Contents

A Proofs from Section 2 (The duality) 2

A.1 Proof of Proposition 1 (Cost function properties) 2
A.2 Proof of Corollary 1 (Shape of G determines curvature) 2
A.3 Proof of Corollary 2 (Canonical distributions) 2
A.4 Proof of Proposition 2 (Universality) 2
A.5 Topological non-robustness of the degenerate case 3
A.6 Proof of Proposition 3 (Dispersion and convexity) 3
A.7 Proof of Proposition 4 (Entropy correspondence) 4
A.8 Detailed entropy discussion 4
A.9 Proof of Proposition 5 (Multi-action duality) 5

B Investment application 6

C Proofs from the pricing section 9

C.1 Proof of Proposition 7 (Stationary distribution) 9
C.2 Proof of Proposition 8 (Selection and the menu cost distribution) . . 10
C.3 Proof of Proposition 9 (Calvo approximation) 10
C.4 Proof of Proposition 10 (Nonlinear Phillips curve) 11
C.5 Proof of Proposition 12 (Monotonicity in γ) 12
C.6 Proof of Proposition 13 (Frequency does not determine non-neutrality) 12
C.7 Proof of Corollary 5 ($\kappa = (1 + 2\gamma)\bar{\Lambda}$ for power laws) 12
C.8 Proof of Corollary 4 (Ordering of non-neutrality) 13

D Supplementary figures 14

E Dominick's moment exercise 14

F Comparison with Williams-Daly-Zachary 17

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A Proofs from Section 2 (The duality)

A.1 Proof of Proposition 1 (Cost function properties)

Proof.
Differentiating $C(\varphi) = w \int_0^{G^{-1}(\varphi)} \xi \, dG(\xi)$: $C'(\varphi) = w\, G^{-1}(\varphi) \cdot g(G^{-1}(\varphi)) \cdot [g(G^{-1}(\varphi))]^{-1} = w\, G^{-1}(\varphi)$, using the inverse function theorem. Differentiating again: $C''(\varphi) = w \cdot [g(G^{-1}(\varphi))]^{-1} > 0$. ■

A.2 Proof of Corollary 1 (Shape of G determines curvature)

Proof.
By Proposition 1(iii), $C''(\varphi) = w/g(G^{-1}(\varphi))$. Since G^{-1} is increasing, g increasing implies $g(G^{-1}(\varphi))$ is increasing in φ, so $C''(\varphi) = w/g(G^{-1}(\varphi))$ is decreasing. The argument for g decreasing and g constant is analogous. ■

A.3 Proof of Corollary 2 (Canonical distributions)

Proof.
Apply $C(\varphi) = w \int_0^{\varphi} G^{-1}(u)\, du$ from Proposition 1(ii). (i) Uniform: $G^{-1}(\varphi) = \bar{\xi}\varphi$, so $C(\varphi) = w\bar{\xi} \int_0^{\varphi} u\, du = \frac{w\bar{\xi}}{2}\varphi^2$. (ii) Power law: $G^{-1}(\varphi) = \bar{\xi}\, \varphi^{1/\gamma}$, so $C(\varphi) = w\bar{\xi} \int_0^{\varphi} u^{1/\gamma}\, du = \frac{\gamma}{\gamma+1} w\bar{\xi}\, \varphi^{1+1/\gamma}$. (iii) Exponential: $G^{-1}(\varphi) = -\lambda^{-1} \ln(1-\varphi)$, so $C(\varphi) = -\frac{w}{\lambda} \int_0^{\varphi} \ln(1-u)\, du = \frac{w}{\lambda}[(1-\varphi)\ln(1-\varphi)+\varphi]$. (iv) Degenerate at $\bar{\xi}$: $G^{-1}(\varphi) = \bar{\xi}$ for all $\varphi \in (0, 1]$, giving $C(\varphi) = w\bar{\xi}\varphi$. ■

A.4 Proof of Proposition 2 (Universality)

Proof.
For $G_\epsilon = U[\bar{\xi} - \epsilon,\ \bar{\xi} + \epsilon]$, the quantile function is $G_\epsilon^{-1}(\varphi) = (\bar{\xi} - \epsilon) + 2\epsilon\varphi$ and the density is $g_\epsilon = 1/(2\epsilon)$. Applying $C(\varphi) = w \int_0^{\varphi} G^{-1}(t)\, dt$:

$$C_\epsilon(\varphi) = w \int_0^{\varphi} G_\epsilon^{-1}(t)\, dt = w \int_0^{\varphi} \big[(\bar{\xi} - \epsilon) + 2\epsilon t\big]\, dt = w(\bar{\xi} - \epsilon)\varphi + w\epsilon\varphi^2.$$

Strict convexity: $C_\epsilon''(\varphi) = 2w\epsilon > 0$ for all $\epsilon > 0$. The first-order condition $S = C_\epsilon'(\varphi) = w(\bar{\xi} - \epsilon) + 2w\epsilon\varphi$ gives $\varphi_\epsilon^*(S) = [S - w(\bar{\xi} - \epsilon)]/(2w\epsilon)$, which is interior whenever S falls in the interval $(w(\bar{\xi} - \epsilon),\ w(\bar{\xi} + \epsilon))$. As $\epsilon \to 0$, this interval collapses to the single point $w\bar{\xi}$, and the interior solution collapses to a step function: φ^* jumps discontinuously from 0 to 1 at $S = w\bar{\xi}$. ■

A.5 Topological non-robustness of the degenerate case

The claim that the lumpy/degenerate case is non-robust can be made precise in the weak topology on the space of probability measures on $[0, \bar{\xi}_{max}]$ for some $\bar{\xi}_{max} > \bar{\xi}$. A sequence of measures μ_n converges weakly to μ if $\int f\, d\mu_n \to \int f\, d\mu$ for every bounded continuous f. The degenerate measure $\delta_{\bar{\xi}}$ (a point mass at $\bar{\xi}$) is the unique element of this space that induces a linear cost function and a binary threshold rule. Yet every weakly open neighbourhood of $\delta_{\bar{\xi}}$ contains measures with continuous positive densities, and hence strictly convex induced costs. To see this, let U be any weakly open neighbourhood of $\delta_{\bar{\xi}}$. By the Portmanteau theorem, U contains all measures μ satisfying $|\int f_i\, d(\mu - \delta_{\bar{\xi}})| < \epsilon$ for finitely many bounded continuous test functions $f_1, \ldots, f_k$. The uniform perturbation $G_\epsilon = U[\bar{\xi} - \epsilon, \bar{\xi} + \epsilon]$ satisfies $\int f_i\, dG_\epsilon \to f_i(\bar{\xi})$ as $\epsilon \to 0$ by continuity, so $G_\epsilon \in U$ for ϵ small enough. Since G_ϵ has a continuous positive density, it induces a strictly convex C_ϵ by Proposition 2. Therefore $\delta_{\bar{\xi}}$ lies in the closure of the set of absolutely continuous measures: the degenerate (lumpy) case is a limit point of the convex (smooth) case, and every neighbourhood of it contains non-degenerate, strictly-convex instances.

A.6 Proof of Proposition 3 (Dispersion and convexity)

Proof.

Part (i): By Proposition 1(i), $C_i(1) = w\, E_{G_i}[\xi]$. Same mean gives $C_1(1) = C_2(1)$.

Part (ii): By Proposition 1(ii), $C_i(\varphi) = w \int_0^\varphi G_i^{-1}(u)\, du$. The standard quantile characterisation of mean-preserving spreads states that G_2 is a MPS of G_1 if and only if

$$\int_0^t G_2^{-1}(u)\, du \le \int_0^t G_1^{-1}(u)\, du \quad \text{for all } t \in [0, 1],$$

with equality at $t = 1$. Therefore $C_1(\varphi) - C_2(\varphi) = w \int_0^\varphi [G_1^{-1}(u) - G_2^{-1}(u)]\, du \ge 0$ for all φ, with equality at $\varphi \in \{0, 1\}$.

Part (iii): Since $C_i'(\varphi) = wG_i^{-1}(\varphi)$ and both quantile functions are continuous and increasing with the same integral over $[0, 1]$, while G_2^{-1} is stretched relative to G_1^{-1} (lower at the bottom, higher at the top), $C_2' - C_1'$ changes sign at least once. For the power law family and other standard parametric families, the crossing is unique; in general, multiple crossings are possible without additional regularity. ■

A.7 Proof of Proposition 4 (Entropy correspondence)

Proof.
Part (i): The quantile function is $G^{-1}(\varphi) = -\lambda^{-1}\ln(1-\varphi)$. Then

$$C(\varphi) = w\int_0^{\varphi} G^{-1}(t)\,dt = -\tfrac{w}{\lambda}\int_0^{\varphi}\ln(1-t)\,dt = \tfrac{w}{\lambda}\big[(1-\varphi)\ln(1-\varphi)+\varphi\big].$$

Parts (ii)–(iii) follow from Corollary 2. ■

A.8 Detailed entropy discussion

Rational inattention and exponential fixed costs Sims (2003) introduces rational inattention, where agents face a Shannon mutual information cost of processing information about a continuous state. Matějka and McKay (2015) show that in the discrete-choice specialisation, this cost reduces to a negative Shannon entropy penalty over the choice probability vector, yielding logit choice probabilities. Proposition 4(i) shows that the cost function induced by exponentially-distributed fixed costs for the binary decision is the KL divergence from the no-adjustment prior — the binary specialisation of the mutual information cost under a degenerate (point-mass) prior on inaction. For binary choice, the two models produce the same adjustment probabilities and the same payoffs. This equivalence does not extend to the full continuous-state RI framework of Sims (2003), where the mutual information cost has a richer structure involving Bayesian updating over a continuous state space. The exact Shannon-entropy form (not just KL divergence) is recovered in the multi-action extension with i.i.d. Gumbel costs (Proposition 5(ii)).

Sparsity and uniform fixed costs Gabaix (2014) penalises attention allocation with a quadratic cost, yielding "sparse" decisions that underreact to stimuli. By Proposition 4(iii), the cost function induced by uniform fixed costs is exactly this quadratic penalty. The equivalence is at the level of the attention cost function for the binary decision; the full sparsity framework includes additional structure (endogenous attention allocation across multiple dimensions) not captured by the binary duality.

Scope of the correspondences The connection to rational inattention is worth making explicit, and stating its limits. Matějka and McKay (2015) show that when an agent faces uncertainty about a state and processes information subject to a Shannon entropy cost, the optimal choice rule takes the multinomial logit form. The standard logit derivation in McFadden (1974) obtains the same form via additive Gumbel *taste shocks* on payoffs. Proposition 5(ii) recovers the multinomial logit by reinterpreting

the cost variable ξ as a Gumbel-distributed payoff shock (allowing negative values, which the bounded-support physical-cost interpretation does not accommodate). At the level of the reduced form, the three setups — RI with Shannon entropy, additive Gumbel payoff shocks, and Gumbel cost shocks under the present convention — yield the same multinomial logit choice probabilities. They are not the same economic model: the underlying primitive (information cost, taste shock, physical cost) and the welfare implications differ.

Within the binary case, exponential G delivers the generalised KL divergence (I-divergence) over the no-adjustment prior, which coincides with the binary specialisation of mutual information under a degenerate prior on inaction. This is a reduced-form identification, not a derivation of RI from physical costs.

A.9 Proof of Proposition 5 (Multi-action duality)

Proof.

Part (i): General case. The agent observes cost draws $(\xi_1, \ldots, \xi_J)$ with $\xi_j \sim G_j$ independently, and selects the action with the highest net payoff:

$$\Pi^{FC}(\mathbf{S}) = E\left[\max_{j\in\{0,\ldots,J\}} \{S_j - w\xi_j\}\right].$$

Partition the expectation by which action is optimal. For each $j \geq 1$, action j is chosen when $S_j - w\xi_j \geq S_k - w\xi_k$ for all $k \neq j$ and $S_j - w\xi_j \geq 0$. Under independence, the probability that action j is chosen can be written as a function $\varphi_j(\mathbf{S})$, and the expected cost paid conditional on choosing j is $E[w\xi_j \mid j \text{ chosen}]$.

Define $\varphi_0 = 1 - \sum_{j=1}^{J} \varphi_j$ (the probability of inaction). The expected payoff is:

$$\Pi^{FC}(\mathbf{S}) = \sum_{j=0}^{J} \varphi_j^* S_j - \sum_{j=1}^{J} E[w\xi_j \cdot \mathbf{1}_{j \text{ chosen}}].$$

By the Fenchel conjugate construction, $\Pi^{FC}(\mathbf{S}) = \sup_{\boldsymbol{\varphi}\in\Delta^J}\{\sum_j \varphi_j S_j - C(\boldsymbol{\varphi})\}$ where $C(\boldsymbol{\varphi})$ is the convex conjugate of Π^{FC}. The conjugate C is jointly defined on Δ^J: even with independent G_j, alternatives interact through the simplex constraint and choice competition, so C is not in general additively separable across actions.

Joint convexity of C follows from the fact that $\Pi^{FC}(\mathbf{S})$, as a pointwise maximum of affine functions of $\mathbf{S}$, is convex in $\mathbf{S}$. The Fenchel biconjugate theorem then implies that C, as the conjugate of Π^{FC}, is convex on Δ^J.

Part (ii): Gumbel payoff-shock analogue. The standard derivation of the multinomial logit (McFadden, 1974) adds i.i.d. Gumbel *taste shocks* ε_j to action-

specific payoffs, giving net payoffs $S_j + \varepsilon_j$ with the inaction option $j = 0$ also receiving a shock. To embed this in the present framework, reinterpret ξ as a generalised payoff shock rather than a physical cost: let $\eta_j \equiv -w\xi_j$ play the role of ε_j, so that η_j takes both signs. With $\eta_j \overset{i.i.d.}{\sim} \text{Gumbel}(0, \sigma)$ for $j = 0, 1, \ldots, J$, the expected maximum of $\{S_j + \eta_j\}$ is the classical log-sum-exp:

$$\Pi^{FC}(\mathbf{S}) = \sigma \ln\left(\sum_{j=0}^{J} e^{S_j/\sigma}\right) + \sigma\gamma_E,$$

where γ_E is the Euler–Mascheroni constant. This is convex in $\mathbf{S}$. The gradient gives the multinomial logit:

$$\varphi_j^* = \frac{\partial \Pi^{FC}}{\partial S_j} = \frac{e^{S_j/\sigma}}{\sum_{k=0}^{J} e^{S_k/\sigma}}.$$

The Fenchel conjugate of the log-sum-exp is the negative Shannon entropy:

$$C(\boldsymbol{\varphi}) = \sigma \sum_{j=0}^{J} \varphi_j \ln \varphi_j,$$

non-positive on the interior of Δ^J. This is the multi-action analogue of the binary correspondence in part (i), but the underlying primitive shifts: in part (i), ξ is a non-negative physical cost; in part (ii), ξ (equivalently, $\eta = -w\xi$) is a real-valued payoff shock, the standard random-utility primitive of McFadden (1974). Negative draws of ξ correspond to favourable taste shocks rather than negative physical costs. ■

Remark 1. *The non-negative physical-cost interpretation of part (i) does not extend directly to the Gumbel case because Gumbel has full real-line support. The honest reading is that Proposition 5(ii) provides an analogue of the standard logit derivation within the convex-conjugate framework, with ξ reinterpreted as a payoff shock. The reduced-form choice probabilities and the conjugate cost coincide with the rational-inattention / WDZ formulations; the primitive does not.*

B Investment application

Consider a firm with capital k choosing investment $I \geq 0$ subject to a random fixed cost $\xi \sim G$. If it adjusts, it pays $w\xi$ and obtains the continuation value at capital $k(1-\delta) + I$. If not, it proceeds with depreciated capital $k(1-\delta)$. The surplus from

investing at the optimal scale is

$$S(k) = \max_{I \geq 0} \left\{ -I + \beta(k(1-\delta) + I)^{\alpha} \right\} - \beta(k(1-\delta))^{\alpha}, \tag{1}$$

where I^* satisfies the first-order condition $1 = \alpha\beta(k(1-\delta)+I^*)^{\alpha-1}$. By Theorem 1, the fixed cost formulation (Definition 1 in the main paper) is equivalent to the probability choice formulation (Definition 2), so the firm's problem is exactly

$$V(k) = \max_{\varphi \in [0,1]} \left\{ \varphi \cdot S(k) - C(\varphi) \right\} + \beta(k(1-\delta))^{\alpha}, \tag{2}$$

with C given by the duality.

Proposition 1 (Conditions for smoothness of C)**.**
$C \in C^2(0,1)$ with $C'' > 0$ on the open unit interval — so that $\varphi^*(S)$ is a C^1 function of the surplus — if and only if:

(i) G is absolutely continuous with density $g > 0$ on $(0, \bar{\xi})$.

(ii) g is continuously differentiable ($g \in C^1$) on $(0, \bar{\xi})$.

Both conditions are automatically satisfied by any G with a C^1, positive density. Smoothness fails when G has atoms (producing kinks in C'), when g has interior zeros (producing points where C'' diverges), or when the support of G is disconnected (producing flat regions in C').

The degenerate case — G a point mass at $\bar{\xi}$ — violates condition (i) and produces the canonical lumpy investment model. C is linear, the only optimal choices are $\varphi \in \{0, 1\}$, and the firm either invests fully or not at all.

Proposition 2 (Investment response and the cost density)**.**
In partial equilibrium, the sensitivity of the aggregate investment probability to a small productivity shock that raises the surplus by dS is

$$\frac{dE[\varphi^*(k)]}{dS} = E\left[\frac{g(G^{-1}(\varphi^*(k)))}{w}\right].$$

This is proportional to the expected density of cost draws at the current threshold, averaged across firms. Economies with higher g at the relevant threshold are more investment-elastic.

Proof.
By Theorem 1, the optimal investment probability is $\varphi^*(k) = G(S(k)/w)$. Differen-

tiating with respect to the surplus:

$$\frac{d\varphi^*}{dS} = \frac{g(S/w)}{w} = \frac{g(G^{-1}(\varphi^*))}{w} = \frac{1}{C''(\varphi^*)},$$

where the last equality uses $C''(\varphi) = w/g(G^{-1}(\varphi))$ from Proposition 1(iii). Taking expectations across the cross-section of firms:

$$\frac{dE[\varphi^*(k)]}{dS} = E\left[\frac{g(G^{-1}(\varphi^*(k)))}{w}\right],$$

which is the expected density at the threshold, averaged over the distribution of capital stocks k. ■

Dispersion as a continuous index of lumpiness The curvature of the induced cost function provides a continuous metric of "how much lumpiness matters." For the family $G = U[\mu - \epsilon, \mu + \epsilon]$, the curvature is $C'' = 2w\epsilon$, strictly increasing in the dispersion ϵ. As $\epsilon \to 0$, the model approaches the binary threshold rule. As ϵ grows, the model resembles a standard convex cost problem. There is no discrete boundary between "lumpy" and "smooth" economies.

Recovering the Khan-Thomas extensive margin under uniform G Khan and Thomas (2008) study a GE model in which firms face i.i.d. fixed costs drawn from a uniform distribution $G = U[0, \bar{\xi}]$. Their central finding is that lumpy micro-level investment has negligible aggregate consequences. The duality makes the mechanism transparent.

Under $G = U[0, \bar{\xi}]$, the duality (equation (2) in the main paper) yields

$$C(\varphi) = w \int_0^{\bar{\xi}\varphi} \xi \cdot \frac{1}{\bar{\xi}}\, d\xi = \frac{w\bar{\xi}}{2}\,\varphi^2.$$

The induced cost is exactly quadratic. Each firm's investment problem therefore becomes

$$V(k) = \max_{\varphi \in [0,1]} \left\{ \varphi \cdot S(k) - \frac{w\bar{\xi}}{2}\,\varphi^2 \right\} + \beta(k(1-\delta))^\alpha,$$

with interior first-order condition $\varphi^*(k) = S(k)/(w\bar{\xi})$. This is a smooth, concave problem — no kinks, no thresholds, no non-convexities. The firm's investment probability responds linearly to the surplus. Aggregating across firms, the total investment rate is a smooth, linear function of the cross-sectional distribution of surpluses. There is nothing "lumpy" left at the aggregate level.

Uniform G is special only in producing *exactly* quadratic costs. More generally, any G with a positive C^1 density produces a strictly convex C^2 cost function (Proposition 1), which already smooths the extensive margin at the firm level.

The power law family $G(\xi) = (\xi/\bar{\xi})^\gamma$ generalises this. When $\gamma = 1$ (uniform), C is quadratic. When $\gamma > 1$ (concentrated G), C is more than quadratic — the cost of inducing high adjustment probabilities rises steeply, producing sharper investment spikes at the micro level. When $\gamma < 1$ (dispersed G), C is less than quadratic — adjustment probabilities respond more elastically to surplus changes. The parameter γ thus indexes the degree of lumpiness along a continuous spectrum, with the Khan and Thomas (2008) quadratic case as the midpoint.

C Proofs from the pricing section

C.1 Proof of Proposition 7 (Stationary distribution)

Proof.
The price gap evolves according to:

$$x_{t+1} = \begin{cases} z_{t+1} & \text{with probability } \Lambda(x_t), \\ x_t + z_{t+1} & \text{with probability } 1 - \Lambda(x_t), \end{cases}$$

where $z_{t+1} \sim \phi_z$ independently. The density of x_{t+1} conditional on the current cross-section f_t is:

$$f_{t+1}(x) = \int \Lambda(\tilde{x}) f_t(\tilde{x})\, d\tilde{x} \cdot \phi_z(x) + \int (1 - \Lambda(\tilde{x}))\phi_z(x - \tilde{x}) f_t(\tilde{x})\, d\tilde{x}.$$

The first term: adjusting firms (total mass $\bar{\Lambda}_t$) draw $z_{t+1} \sim \phi_z$ as their new gap. The second term: non-adjusting firms at $\tilde{x}$ receive shock z_{t+1} and move to $\tilde{x} + z_{t+1}$; the density at x is the convolution of $(1 - \Lambda(\tilde{x}))f_t(\tilde{x})$ with ϕ_z.

Setting $f_{t+1} = f_t = f_{ss}$ yields the fixed-point equation of the main paper. Symmetry: if ϕ_z is symmetric around zero and $\Lambda(x) = \Lambda(-x)$ (which holds since $\Lambda(x) = G(Bx^2/w)$), then if f_t is symmetric, so is f_{t+1}. Since f_{ss} is the unique fixed point of this contraction, f_{ss} is symmetric around zero. ■

Remark 2. *The fixed-point equation corresponds, in discrete state space, to the eigenvector problem* $\mathbf{f} = \mathbf{T}'\mathbf{f}$ *where* $\mathbf{T}$ *is the transition matrix encoding the adjustment and drift probabilities. This is the object computed numerically in the quantitative implementation.*

C.2 Proof of Proposition 8 (Selection and the menu cost distribution)

Proof.
Write the adjustment-weighted mean absolute gap as

$$\overline{|x|}^{adj} = \frac{E[\Lambda(x) \cdot |x|]}{E[\Lambda(x)]} = E[|x|] + \frac{\mathrm{Cov}(\Lambda(x)/\bar{\Lambda},\ |x|)}{1}.$$

The covariance is positive when Λ is increasing in $|x|$. Its magnitude depends on how sharply Λ responds to $|x|$.

(i) The elasticity of the hazard with respect to $|x|$ is

$$\frac{\partial \ln \Lambda(x)}{\partial \ln |x|} = \frac{g(Bx^2/w)}{G(Bx^2/w)} \cdot \frac{2Bx^2}{w},$$

which is twice the hazard rate of G evaluated at the firm's surplus. A higher hazard rate means Λ responds more sharply to $|x|$, increasing $\mathcal{S}$.

(ii) The elasticity in (i) is proportional to g/G, the reversed hazard rate. When g/G is larger at the threshold, the hazard Λ is more responsive to x^2, concentrating adjustment among high-gap firms. When g/G is small, Λ is nearly flat and selection is weak.

(iii) For constant Λ (Calvo), the covariance is zero and $\mathcal{S} = 0$. For a step function Λ (Golosov-Lucas), $\mathrm{Cov}(\Lambda/\bar{\Lambda}, |x|)$ is at the polar extreme of the power law family at fixed $\bar{\Lambda}$, attaining the largest $\mathcal{S}$ within that family. A general extremality statement across all hazards calibrated to the same $\bar{\Lambda}$ would require a rearrangement-style argument not pursued here.

For the power law family, the hazard rate of G at ξ is $g(\xi)/G(\xi) = \gamma/\xi$, so the elasticity is 2γ — constant and strictly increasing in γ. ■

C.3 Proof of Proposition 9 (Calvo approximation)

Proof.
For the power law family $G(\xi) = (\xi/\bar{\xi})^\gamma$ with $\bar{\xi}(\gamma)$ calibrated so that $\bar{\Lambda} = \lambda$, Corollary 5 gives $\kappa = (1 + 2\gamma)\lambda$ (provided f_{ss} is supported in the interior of the inaction band).

(i) The Phillips curve slope is $\kappa = (1 + 2\gamma)\lambda \to \lambda$ as $\gamma \to 0^+$.

(ii) Non-neutrality is $1 - \kappa = 1 - (1 + 2\gamma)\lambda \to 1 - \lambda$ as $\gamma \to 0^+$.

Remark on the nature of the limit. The hazard $\Lambda(x) = (Bx^2/(w\bar{\xi}))^\gamma$ does not become constant as $\gamma \to 0^+$. With fixed $\bar{\xi}$, $\Lambda(x) \to 1$ for any $x \neq 0$ (near-certain adjustment,

not Calvo). The Calvo approximation arises because, in the frequency-calibrated limit, $\bar{\xi}(\gamma) \to \infty$ as $\gamma \to 0^+$, pushing the inaction band outward so that the hazard is small and approximately linear in x^2 over the support of f_{ss}. The hazard remains state-dependent at every $\gamma > 0$; what vanishes is the *aggregate consequence* of this state dependence, as measured by the selection wedge $2\gamma\lambda$. ■

C.4 Proof of Proposition 10 (Nonlinear Phillips curve)

Proof.
Assume that f_{ss} is continuously differentiable with $\int |x|\, f_{ss}(x)\, dx < \infty$ and that G is C^1 with bounded density g, so that differentiation under the integral sign is justified by the dominated convergence theorem. Write $f_t(x) = f_{ss}(x - \mu_t)$. Then

$$\pi_t = \int G\left(\frac{Bx^2}{w}\right) x \cdot f_{ss}(x - \mu_t)\, dx.$$

Differentiating with respect to μ_t using Leibniz's rule:

$$\frac{\partial \pi_t}{\partial \mu_t} = -\int G\left(\frac{Bx^2}{w}\right) x \cdot f'_{ss}(x - \mu_t)\, dx,$$

since $\frac{\partial}{\partial \mu_t} f_{ss}(x - \mu_t) = -f'_{ss}(x - \mu_t)$. Evaluating at $\mu_t = 0$:

$$\left.\frac{\partial \pi_t}{\partial \mu_t}\right|_{\mu_t=0} = -\int G\left(\frac{Bx^2}{w}\right) x \cdot f'_{ss}(x)\, dx.$$

Now integrate by parts with $u = G(Bx^2/w) \cdot x$ and $dv = f'_{ss}(x)\, dx$:

$$-\int u\, dv = -\left[G\left(\tfrac{Bx^2}{w}\right) x \cdot f_{ss}(x)\right]_{-\infty}^{\infty} + \int f_{ss}(x)\, \frac{d}{dx}\left[G\left(\tfrac{Bx^2}{w}\right) x\right] dx.$$

The boundary term vanishes because f_{ss} has tails decaying faster than any polynomial. Expanding the derivative:

$$\frac{d}{dx}\left[G\left(\frac{Bx^2}{w}\right) x\right] = g\left(\frac{Bx^2}{w}\right) \frac{2Bx}{w} \cdot x + G\left(\frac{Bx^2}{w}\right) = g\left(\frac{Bx^2}{w}\right) \frac{2Bx^2}{w} + G\left(\frac{Bx^2}{w}\right).$$

Substituting:

$$\kappa(G) \;=\; \left.\frac{\partial \pi_t}{\partial \mu_t}\right|_{\mu_t=0} \;=\; \int G\left(\frac{Bx^2}{w}\right) f_{ss}(x)\, dx + \int g\left(\frac{Bx^2}{w}\right) \frac{2Bx^2}{w}\, f_{ss}(x)\, dx.$$

The first integral is the extensive margin. The second integral is the intensive margin. Since $G(Bx^2/w)$ is even in x and f_{ss} is symmetric around zero, $\pi = 0$ at $\mu_t = 0$; the linearisation coefficient κ is well-defined and given by the formula above. ■

C.5 Proof of Proposition 12 (Monotonicity in γ)

Proof.
Under the interior-support hypothesis, Corollary 5 gives $\kappa = (1 + 2\gamma)\bar{\Lambda}$ and selection wedge $2\gamma\bar{\Lambda}$ as exact equalities; the monotonicity statements below are immediate at any $\bar{\Lambda} > 0$. Under typical normal-shock calibrations the hypothesis holds approximately, and the same monotonicities hold to that approximation.

(i) The normalised selection wedge is $(\kappa - \bar{\Lambda})/\bar{\Lambda} = 2\gamma$, strictly increasing in γ.

(ii) Monetary non-neutrality is proportional to $1 - \kappa = 1 - (1 + 2\gamma)\bar{\Lambda}$, strictly decreasing in γ for any $\bar{\Lambda} > 0$.

(iii) The Phillips curve slope is $\kappa = (1 + 2\gamma)\bar{\Lambda}$, strictly increasing in γ.

In each case, calibrating two power law economies to the same $\bar{\Lambda}$ (by adjusting $\bar{\xi}$) preserves γ as the sole determinant of κ, $\mathcal{S}$, and non-neutrality. ■

Remark 3. *Proposition 12 is stated within the power law family, which provides a clean one-parameter ordering at equal frequency. The underlying mechanism — concentrated G steepens the adjustment hazard, amplifying selection — is not family-specific. A general monotonicity result would require a stochastic-order condition on G that ensures higher $\kappa/\bar{\Lambda}$ whenever G is "more concentrated" in a suitable sense.*

C.6 Proof of Proposition 13 (Frequency does not determine non-neutrality)

Proof.
By the Phillips curve slope formula, $\bar{\Lambda}$ depends on $E[G(Bx^2/w)]$ while κ depends additionally on $E[g(Bx^2/w) \cdot 2Bx^2/w]$. Matching the first does not determine the second. For the power law $G(\xi) = (\xi/\bar{\xi})^\gamma$, direct computation gives $g(Bx^2/w) \cdot 2Bx^2/w = 2\gamma\,\Lambda(x)$, so $\kappa = (1 + 2\gamma)\bar{\Lambda}$ under the interior-support condition (Corollary 5). Two power laws with different γ, calibrated to the same $\bar{\Lambda}$ by adjusting $\bar{\xi}$ and with f_{ss} chosen to have support inside the inaction band, produce different κ. ■

C.7 Proof of Corollary 5 ($\kappa = (1 + 2\gamma)\bar{\Lambda}$ for power laws)

Proof.
For $G(\xi) = (\xi/\bar{\xi})^\gamma$, the density is $g(\xi) = (\gamma/\bar{\xi})(\xi/\bar{\xi})^{\gamma-1}$. Evaluate the intensive margin

integrand at $\xi = Bx^2/w$:

$$
\begin{aligned}
g\left(\frac{Bx^2}{w}\right) \cdot \frac{2Bx^2}{w} &= \frac{\gamma}{\bar{\xi}} \left(\frac{Bx^2/w}{\bar{\xi}}\right)^{\gamma-1} \cdot \frac{2Bx^2}{w} \\
&= 2\gamma \cdot \frac{1}{\bar{\xi}} \left(\frac{Bx^2/w}{\bar{\xi}}\right)^{\gamma-1} \cdot \frac{Bx^2}{w} \\
&= 2\gamma \left(\frac{Bx^2}{w\bar{\xi}}\right)^{\gamma} = 2\gamma\, \Lambda(x).
\end{aligned}
$$

Substituting into the Phillips curve slope formula:

$$
\begin{aligned}
\kappa(G) &= \int G\left(\frac{Bx^2}{w}\right) f_{ss}(x)\, dx + \int 2\gamma\, \Lambda(x)\, f_{ss}(x)\, dx \\
&= \bar{\Lambda} + 2\gamma\bar{\Lambda} = (1+2\gamma)\bar{\Lambda}.
\end{aligned}
$$

The selection wedge is $\kappa - \bar{\Lambda} = 2\gamma\bar{\Lambda}$. The ratio $\kappa/\bar{\Lambda} = 1 + 2\gamma$ depends only on γ, not on f_{ss}, B, or w. The identity requires that f_{ss} is supported in the interior of the inaction band ($|x| < \sqrt{w\bar{\xi}/B}$ f_{ss}-a.s.), so that the uncapped power law expression $\Lambda(x) = (Bx^2/(w\bar{\xi}))^{\gamma} < 1$ applies throughout. ■

C.8 Proof of Corollary 4 (Ordering of non-neutrality)

Proof.
The real effect of a monetary shock dm is proportional to $(1-\kappa)\, dm$, so non-neutrality is decreasing in κ. The argument below restricts to the power law family $G(\xi) = (\xi/\bar{\xi})^{\gamma}$, where the closed-form $\kappa = (1+2\gamma)\bar{\Lambda}$ of Corollary 5 makes the ordering exact.

(i) Calvo benchmark ($\gamma \to 0^+$). Within the power law family, the frequency-calibrated limit $\gamma \to 0^+$ (with $\bar{\xi}(\gamma) \to \infty$ to hold $\bar{\Lambda}$ fixed; Proposition 9) yields $\kappa - \bar{\Lambda} = 2\gamma\bar{\Lambda} \to 0$. The selection wedge vanishes, and non-neutrality $1 - \kappa \to 1 - \bar{\Lambda}$, the same-frequency Calvo level.

(ii) Interior $\gamma > 0$. By Corollary 5, the selection wedge equals $2\gamma\bar{\Lambda} > 0$ and is strictly increasing in γ. Non-neutrality $1 - \kappa = 1 - (1+2\gamma)\bar{\Lambda}$ is strictly decreasing in γ (provided $\kappa < 1$).

(iii) Golosov-Lucas ($\gamma \to \infty$). The hazard converges to a step function at the inaction-band boundary, and the selection wedge attains its supremum within the power law family. Non-neutrality is minimised within the family.

A general statement covering all absolutely continuous G (e.g., that Calvo achieves the maximum non-neutrality among same-frequency state-dependent hazards, or that Golosov-Lucas achieves the minimum) requires a rearrangement-style argument that

we do not pursue here; the corollary as stated in the main text is restricted to the power law family. ■

D Supplementary figures

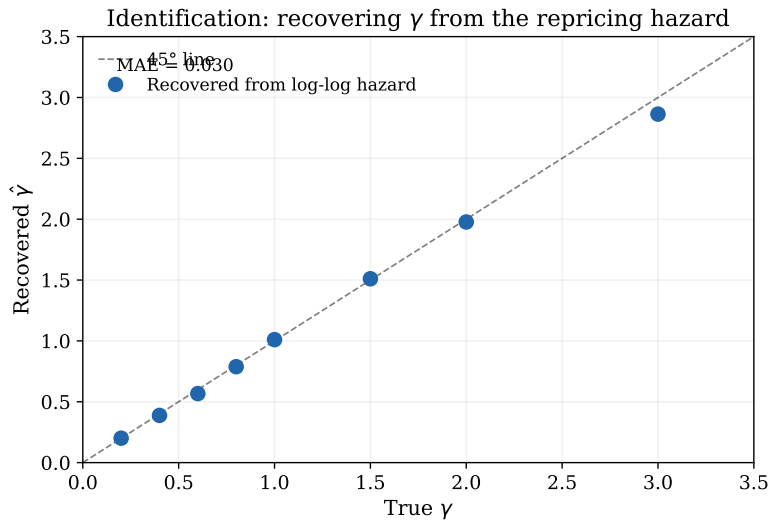

Figure 1: Recovery of γ from the log-log hazard regression. Each point is one simulated economy with 15,000 firms and 150 quarters. The regression recovers the true γ with mean absolute error 0.012. The identification requires only the conditional adjustment probability as a function of the price gap.

E Dominick's moment exercise

This appendix runs a moment exercise on the Dominick's Finer Foods scanner dataset (Midrigan, 2011; Chevalier, Kashyap, and Rossi, 2003) to see what additional information the scanner moments provide relative to the CPI-based calibration of Section 4.3 of the main paper. The dataset covers seven product categories (analgesics, cereals, cheeses, cookies, canned soup, laundry detergent, soft drinks) across approximately 100 stores and 400 weeks, yielding 42 million store-week-UPC observations. Temporary sales are removed using both the recorded sale indicator and a V-shaped filter that flags price drops reversed within three weeks.

Moments Table 1 reports the key moments. The kurtosis of non-sale price changes is 7.6, and 21% of changes are smaller than 2.5% in absolute value — both indicating

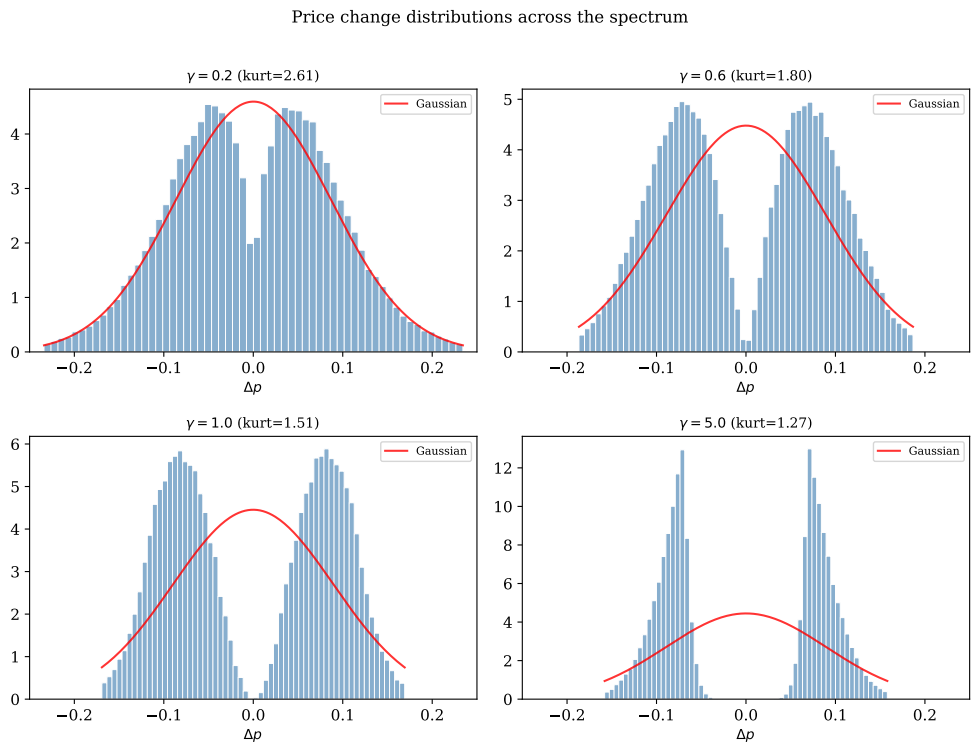

Figure 2: Distribution of price changes (heterogeneous σ_z, CV = 1). Red line: Gaussian with the same mean and variance. Low γ produces excess kurtosis through weak selection (small and large changes coexist). High γ concentrates changes through strong selection.

substantial leptokurtosis and a large mass of small adjustments, consistent with weak selection.

Table 1: Dominick's price change moments (excl. sales)

Moment	**Value**
Frequency (weekly)	9.3%
Frequency (quarterly)	71.8%
Mean $\lvert\Delta p\rvert$	13.5%
Kurtosis	7.62
Fraction $\lvert\Delta p\rvert < 2.5\%$	21%
Fraction $\lvert\Delta p\rvert < 5\%$	42%
N (price changes, excl. sales)	1,656,025

Moment-based identification The Dominick's quarterly repricing frequency (0.72) is considerably higher than the CPI-based target ($\bar{\Lambda} = 0.24$). The baseline simulated-moments table of the main paper is therefore not directly applicable; the model must be re-calibrated to Dominick's frequency before matching the other moments. Table 2 reports the recalibrated heterogeneous-σ_z moments.

Table 2: Heterogeneous-σ_z moments at Dominick's frequency

γ	$\bar{\xi}$	**Kurtosis**	**Frac** $< 2.5\%$
0.10	0.0160	7.67	41%
0.15	0.0047	7.64	39%
0.20	0.0024	7.53	37%
0.25	0.0015	7.63	36%
0.30	0.0011	7.68	35%
0.40	0.0007	7.51	34%
Data	—	7.62	21%

Notes: Heterogeneous-σ_z simulation (CV = 1) with $\bar{\xi}$ calibrated for each γ to match $\bar{\Lambda} = 0.72$ quarterly. 8000 firms, 80 quarters after 40 quarters burn-in.

Two features of Table 2 stand out. First, at the Dominick's frequency, simulated kurtosis is essentially flat across $\gamma \in [0.10, 0.40]$, matching the empirical value of 7.62 for every γ in this range. Kurtosis alone therefore does not identify γ sharply at this frequency: it bounds γ below ~ 0.4 but does not tie it down further. Second, the empirical fraction of small price changes (21%) is *lower* than the model generates at any γ in this range. This is a model-fit limitation — at Dominick's frequency, the heterogeneous-σ_z specification produces too many small price changes, and the power law family cannot close the gap without additional structure (e.g., richer cross-product heterogeneity in G). The fragility contrasts with the CPI-based calibration of Section 4.3 of the main paper, where $\bar{\Lambda} = 0.24$ leaves more of the heterogeneity-driven kurtosis variation visible, so kurtosis selects a central calibrated value near $\hat{\gamma} \approx 0.4$. The difference reflects the underlying difference in data: scanner data from a single grocery chain exhibits much higher repricing frequency, more product-level heterogeneity, and more small price changes than the broad CPI coverage underlying the Nakamura and Steinsson (2010) moments. The reconciliation suggests that G varies across sectors. Oskolkov and Lippi (2026) provide direct evidence of this in plant-level investment data. Modelling sectoral heterogeneity explicitly — with sector-specific $(\gamma_s, \bar{\xi}_s)$ drawn from a meta-distribution — is a natural next step that would discipline the pooled calibration, and would also test whether the CPI-based $\hat{\gamma} \approx 0.4$ is best read as a population parameter or as an aggregation of sectoral γ's.

Distribution of price changes Figure 3 shows the distribution of non-sale price changes: strongly leptokurtic, with a sharp peak near zero and heavy tails — qualitatively matching the model's prediction for low γ.

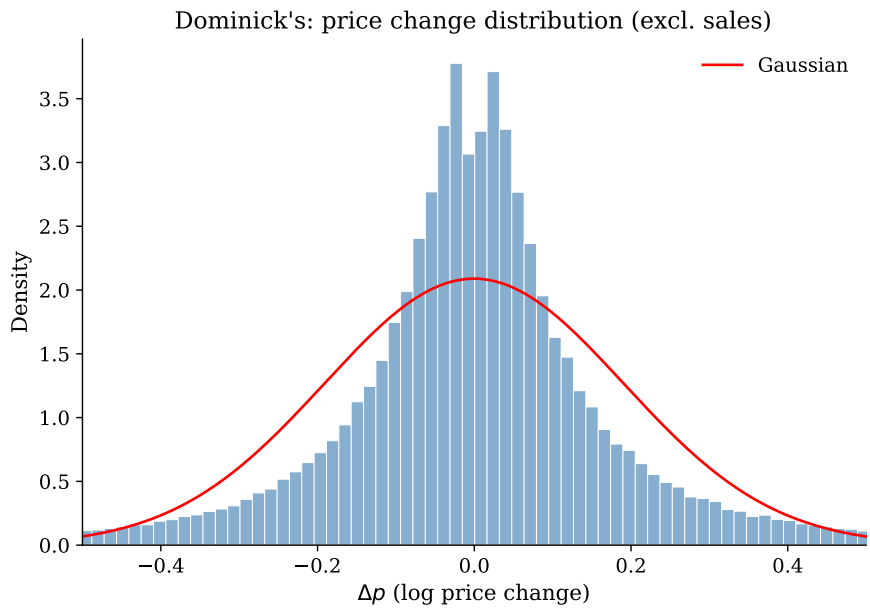

Figure 3: Distribution of non-sale price changes in Dominick's

Notes: 7 categories, 42 million observations. The distribution is strongly leptokurtic (kurtosis = 7.6), with a sharp peak near zero and heavy tails relative to the Gaussian benchmark (red). Consistent with the model's prediction for low γ.

The duration-based repricing hazard is steeply decreasing — negative duration dependence documented by Nakamura and Steinsson (2008). Within the present framework, this is a signature of cross-product heterogeneity in G: the within-product hazard $\Lambda(x) = G(Bx^2/w)$ is increasing in $|x|$ for each product, but the pooled hazard across products with different G's can be decreasing in duration. Negative duration dependence in the aggregate is consistent with positive state dependence at the product level, provided menu costs are heterogeneous across products.

F Comparison with Williams-Daly-Zachary

Section 2.2 of the main paper discusses the relationship between Theorem 1 and the Williams-Daly-Zachary (WDZ) theorem (McFadden, 1978, 1981). This appendix provides a systematic comparison. Both results share a common mathematical skeleton

— an expected-maximum operator equals the Fenchel conjugate of a function determined by a distributional primitive — but differ in their economic content, scope, and implications. Table 3 summarises the comparison along six dimensions.

Table 3: Williams-Daly-Zachary vs. Theorem 1

Dimension	**WDZ**	**Theorem 1**
Random element	Taste shocks ε_j on payoffs	Cost draws $w\xi$ on adjustment
Decision	Multi-alternative discrete choice	Binary: adjust or not
Surplus function	$E[\max_j\{u_j + \varepsilon_j\}]$	$\Pi^{FC}(S)$ (Definition 1)
Convexity result	Surplus convex in $(u_1, \ldots, u_J)$	Payoff convex in S
Conjugate of	Generating function $\mathcal{G}(\varepsilon)$	$C(\varphi) = \int_0^\varphi G^{-1}(t)\,dt$
Choice probs	$p_j = \partial E[\max]/\partial u_j$	$\varphi^* = G(S)$
Primitive	Joint distribution of ε	Distribution G of costs
Coupling	Not constructed	Path-by-path equivalence
Economic context	Demand: consumer choice	Supply: adjustment frictions
Application	Multinomial logit (McFadden, 1974)	Calvo–GL spectrum

For binary choice ($J = 2$), the two results are formally equivalent under a sign change: a random benefit $S - w\xi$ versus a random cost $w\xi$ applied to a fixed benefit S are isomorphic. The binary conjugate identity is therefore not new. The contributions of this paper lie beyond the duality identity itself: the mapping $G \mapsto C$ and the distributional taxonomy (Propositions 3–5), the structural explanation for the Khan and Thomas (2008) result, and the continuous parametrisation of the pricing spectrum are entirely absent from WDZ and the discrete choice literature.

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